Statistics & Sampling

ANSWER KEY

Section 1: Measures of Center & Spread

1 A data set consists of the numbers $4, 7, 8, 11, 14, 16, 24$. What is the median of the data set?

a. 8

b. 11 ✓

c. 12

d. 14

With 7 ordered values, the median is the 4th value: 11.

2 A student scored $82, 88, 91, 85$ on four tests. What score must the student earn on the fifth test to have an overall mean of 88?

a. 90

b. 92

c. 94 ✓

d. 96

Total points required for mean of 88 over 5 tests is $5 \times 88 = 440$. Sum of first 4 tests: $82 + 88 + 91 + 85 = 346$. Fifth score required: $440 - 346 = 94$.

3 If an extreme outlier of 150 is added to a data set with a mean of 45 and a median of 42, which statistic will increase by the greatest amount?

a. The median

b. **The mean** ✓

c. The mode

d. The standard deviation will remain constant

The mean is sensitive to extreme outliers because every value is factored into the sum, whereas the median is resistant.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Section 2: Range & Standard Deviation

4 Data Set A has values $10, 10, 10, 10$. Data Set B has values $5, 10, 10, 15$. Which statement is true regarding their standard deviations?

a. **The standard deviation of Data Set A is 0, which is less than that of Data Set B.** ✓

b. The standard deviation of Data Set A is equal to that of Data Set B.

c. Data Set A has a greater standard deviation than Data Set B.

d. Neither data set has a defined standard deviation.

Data Set A has zero variability (all identical), so its standard deviation is 0. Data Set B has spread around 10, so its standard deviation is positive.

5 The range of a set of seven test scores is 28. If the lowest score in the set is 64, what is the highest score?

a. 88

b. 90

c. 92 ✓

d. 96

$\text{Range} = \text{Max} - \text{Min} \Rightarrow 28 = \text{Max} - 64 \Rightarrow \text{Max} = 64 + 28 = 92$.

6 If every value in a dataset is multiplied by 3, how are the mean and standard deviation affected?

a. **The mean triples, and the standard deviation triples.** ✓

b. The mean triples, but the standard deviation remains unchanged.

c. The mean remains unchanged, but the standard deviation triples.

d. Both the mean and standard deviation increase by 3.

Multiplying every observation by a constant $c = 3$ scales both the measures of center (mean) and measures of spread (standard deviation) by $|c| = 3$.

Section 3: Sampling Methods & Bias

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 A researcher surveys the first 50 people entering a fitness center at 6:00 AM regarding the average number of hours they exercise per week. To which population can the results be generalized?

a. All adults in the city.

b. All members of that fitness center.

c. **Only patrons who visit the fitness center early in the morning.** ✓

d. All professional athletes.

A convenience sample of early morning visitors introduces bias and can only be generalized to members who visit the gym at that specific early time.

8 Which sampling method produces the most representative, unbiased sample of high school students' opinions on cafeteria food?

a. Surveying students sitting at the first three tables in the cafeteria.

b. **Surveying 100 students chosen at random from an alphabetical student roster.** ✓

c. Posting an online poll and asking volunteers to reply.

d. Surveying only members of the culinary club.

Simple random sampling from the complete school roster ensures every student has an equal chance of selection, eliminating systematic selection bias.

9 A randomized controlled experiment is conducted where participants are randomly assigned to either a new memory technique group or a control group. What does random assignment allow the researchers to establish?

a. **A causal relationship between the technique and memory improvement.** ✓

b. That the sample represents the entire global population.

c. That no participants experienced any variation.

d. That the margin of error is zero.

Random assignment of subjects to treatments controls for confounding variables and allows determination of cause-and-effect (causation).

Section 4: Margin of Error & Confidence

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 A random sample poll reports that 54% of voters support a bond proposal with a margin of error of $\pm 3.5\%$. Which interval is plausible for the true population percentage?

a. 50.5% **to** 57.5% ✓

b. 52.0% to 56.0%

c. 47.0% to 61.0%

d. 54.0% to 57.5%

Confidence interval is $\hat{p} \pm \text{MOE} = 54\% \pm 3.5\% = [50.5\%, 57.5\%]$.

11 If a polling organization increases its random sample size from 400 to $1,600$ respondents, how will the margin of error change?

a. **It will be halved.** ✓

b. It will be quadrupled.

c. It will be reduced to one-fourth.

d. It will remain unchanged.

The margin of error is inversely proportional to the square root of sample size: $\frac{1}{\sqrt{1600}} / \frac{1}{\sqrt{400}} = \frac{20}{40} = \frac{1}{2}$. It is cut in half.

12 A study estimates that the mean weight of newborn calves is 78 pounds with a margin of error of 4 pounds at a 95% confidence level. Which of the following is the most accurate conclusion?

a. **It is plausible that the true population mean weight is between** 74 **and** 82 **pounds.** ✓

b. Exactly 95% of all calves weigh between 74 and 82 pounds.

c. The weight of every calf in the population is at least 74 pounds.

d. The sample mean was between 74 and 82 pounds.

A margin of error around the sample mean provides an interval of plausible values for the true population parameter ($78 \pm 4 = [74, 82]$ pounds).

© ClipSAT: By Mr. Mohamed Abdallah +966597688647